

Problem 9

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Problem 1 Find the domain of the function. Determine whether the function is odd, even, or neither.

$$(a) f(x) = \frac{x}{\sqrt{x^2 - 9}}$$

Explanation. To find the domain, recall that a rational expression cannot have 0 in the denominator and a square root expression cannot have a negative number under the square root. Thus, $x^2 - 9 > 0 \implies (x - 3)(x + 3) > 0$. The zeros are located at $x = -3, 3$. From this we can draw a sign chart for the expression, $x^2 - 9$, and test values.

alt=sign-chart showing that it is positive when $x < -3$ or when $x > 3$, but is negative if $-3 < x < 3$.

We see that $x^2 - 9 > 0$ on the interval $(-\infty, -3) \cup (3, \infty)$. Thus our domain is $(-\infty, -3) \cup (3, \infty)$.

Next we check for even/odd/neither.

$$f(-x) = \frac{-x}{\sqrt{(-x)^2 - 9}} \text{ which does not equal } f(x) \text{ so } f \text{ is not even}$$

$$-f(x) = -\frac{x}{\sqrt{x^2 - 9}} \text{ which is equal to } f(-x) \text{ so } f \text{ is odd.}$$

$$(b) g(x) = \frac{\sin(x)}{x}$$

Explanation. To find the domain, recall that a rational expression cannot have 0 in the denominator. Therefore, our domain is $(-\infty, 0) \cup (0, \infty)$.

Next we check for even/odd/neither.

$$f(-x) = \frac{\sin(-x)}{-x} = \frac{-\sin(x)}{-x} = \frac{\sin(x)}{x} \text{ which equals } f(x) \text{ so } f \text{ is even}$$

$$(c) h(t) = \ln(t^3 - 1)$$

Explanation. To find the domain, recall that we cannot take the natural logarithm of 0 or a negative number. Therefore, $t^3 - 1 > 0 \implies (t - 1)(t^2 + t + 1) > 0$. The zero is located at $t = 1$. From this we can draw a sign chart and test values.



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We see that $t^3 - 1 > 0$ on the interval $(1, \infty)$. Thus our domain is $(1, \infty)$.

Next we check for even/odd/neither.

$h(t)$ is neither even nor odd because if t is in the domain, then $-t$ is not in the domain.